ITUH USED 27-10-2020PM PAP

Code No: 117EF **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, October/November - 2020
MACHINE LEARNING
(Computer Science and Engineering)

Time: 2 Hours **Max. Marks: 75**

Answer any Five Questions
All Questions Carry Equal Marks
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1. Explain the sequence of design choices made for checkers program in detail with illustrations. [15]

2. Discuss the limitations of FIND-S algorithm and how Candidate Elimination algorithm addresses these limitations. [15]

3. Examine the hypothesis space search performed by ID3 algorithm and contrast it with List and then eliminate algorithm. [15]

4. Describe the derivation of the gradient descent rule. [15]

5.a) How does the deviation between sample error and true error depend on the size of the data sample?
b) Discuss central limit theorem. [8+7]

6. Explain how to use Bayesian classifier to learn text classification with an example.[15]

7. Describe k-nearest neighbor algorithm for approximating a discrete-valued function.[15]

8.a) What is explanation based learning? Compare it with instance-based learning.
b) Demonstrate usage of prior knowledge to reduce the sample complexity. [8+7]

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